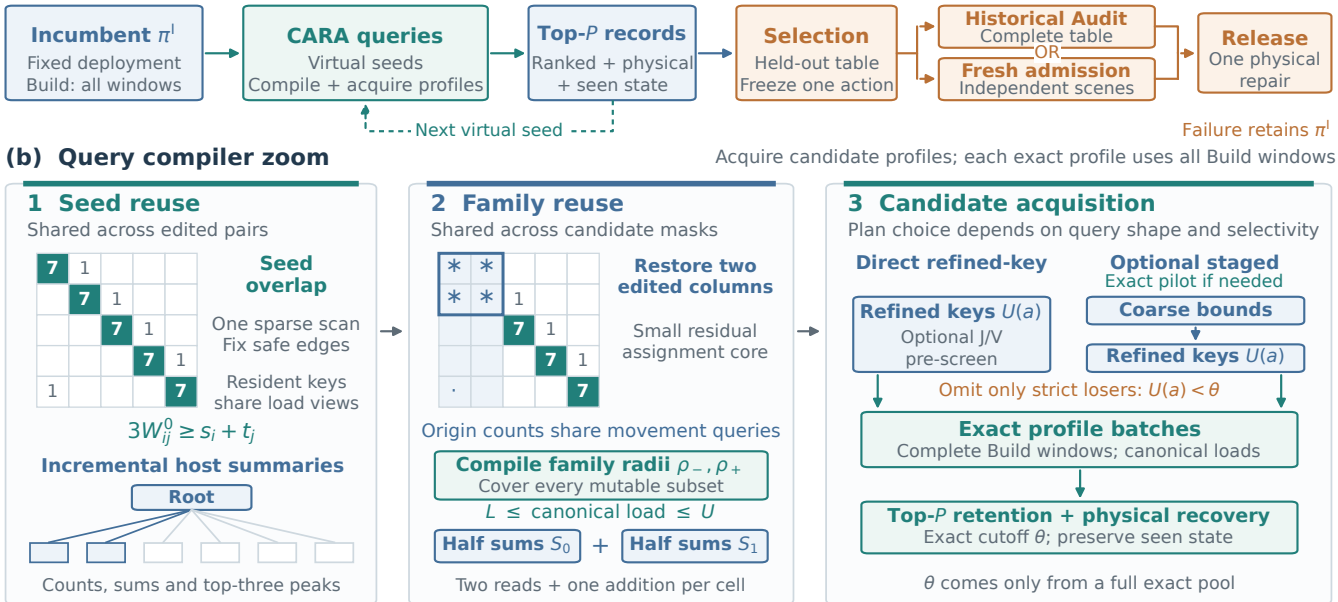
(a) Decision path: many query alternatives, one released repair
Incumbent π^{I}
Fixed deployment
Build: all windows
CARA queries
Virtual seeds
Compile + acquire profiles
Top-P records
Ranked + physical
+ seen state
Next virtual seed
Selection
Held-out table
Freeze one action
Historical Audit
Complete table
OR
Fresh admission
Independent scenes
Release
One physical
repair
Failure retains π^{I}
(b) Query compiler zoom
Acquire candidate profiles; each exact profile uses all Build windows
1 Seed reuse
Shared across edited pairs
Seed overlap
One sparse scan
Fix safe edges
Resident keys
share load views
$3W_{ij}^{0} \geq s_i + t_j$
Incremental host summaries
Root
Counts, sums and top-three peaks
2 Family reuse
Shared across candidate masks
Restore two edited columns
Small residual
assignment core
Origin counts share movement queries
Compile family radii ρ_-, ρ_+
Cover every mutable subset
$L \leq$ canonical load $\leq U$
Half sums S_0 + Half sums S_1
Two reads + one addition per cell
3 Candidate acquisition
Plan choice depends on query shape and selectivity
Direct refined-key
Optional staged
Exact pilot if needed
Refined keys $U(a)$
Optional J/V
pre-screen
Coarse bounds
Refined keys $U(a)$
Omit only strict losers: $U(a) < \theta$
Exact profile batches
Complete Build windows; canonical loads
Top-P retention + physical recovery
Exact cutoff θ; preserve seen state
θ comes only from a full exact pool
Query contract: identical ordered records, canonical profiles, deterministic physical ties and continuing seen state